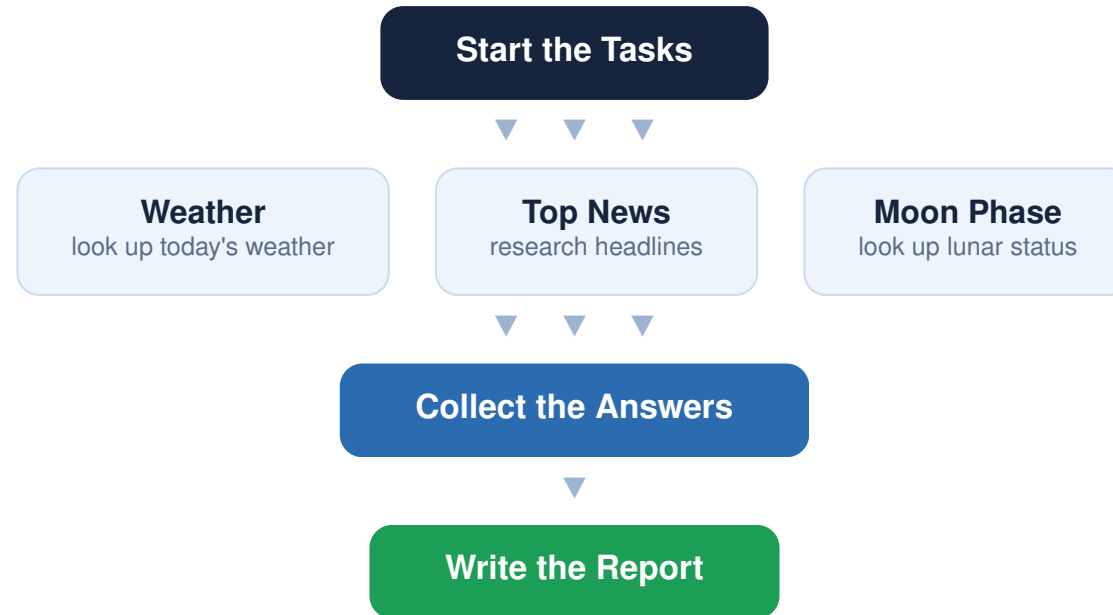


THE FAN-OUT / FAN-IN PROBLEM — EXPLAINED MORE CLEARLY

AI Concurrency: Fan-Out / Fan-In

It is a bit technical — but enlightening. It will speed your reports. And once you know what you want, you can consider running your research **outside** the LLM entirely.

Independent tasks can run at the same time



Because these tasks **do not depend on each other**, you can run them at the same time — started from one program. This saves you time, and puts a bit of stress out on the provider (think ChatGPT or Claude). In practice: **sometimes a provider ran them concurrently, sometimes not** — an improved prompt better encouraged the provider to run them at the same time and collect the answers.

Minimize time to accurately complete a task, at a good price

Time

Parallel tasks finish together instead of waiting on each other one at a time. The wall-clock win is real.

Accuracy

Fan-in composes the final answer only after everything comes back — nothing is skipped or half-done.

Price

Same total work — just rearranged. But watch the per-run cost before you make it a habit.

Keep this in mind for your **personal productivity** — and more importantly, for **your team**. Not every job needs it: fan-out only pays off when the tasks are truly independent.

3 · CHOOSE THE MODEL

Find the smallest model that does the job

\$0.15 – \$0.35

per run, depending on model and exact task
(Claude's estimate)

Haiku < Sonnet

a smaller model can cut the cost of a run roughly
in half

Jelly Beans

on subscription plans, each run takes a chunk out
of your daily (or weekly) allowance

These little helper tasks can be **more expensive than you think**. Consider the cost to you first — then multiply it across the team. The model choice is where the money is.

Each request is small — but they add up

- ▶ **Experimented with Haiku and Claude** to see how model choice changed cost and quality.
- ▶ **Wrote a Python 3 program** to run the research directly, outside the LLM.
- ▶ **Published it as a GitHub project** so others can try it — instead of just reading a post about it.
- ▶ **The prompt is readable right on GitHub** — no need to download the whole project.

GitHub: LLM-FanOut-FanIn

A working example of the fan-out / fan-in pattern — clone it, run it, or just read the prompt in the browser.

The program runs faster — and can be more reliable

By the way — that example program is **not even concurrent**, and it still runs faster than the LLM. When you know exactly what you want, moving the research out of the model and into a small program buys you speed, reliability, and a much smaller bill.

The takeaway: Fan-out / fan-in isn't just faster — it's the same total work, done in parallel instead of in sequence. Use it when the tasks are independent, pick the smallest model that does the job, and when the task is fully understood — write the program.